

Spatially Explicit Rock–TBM Interaction Graph Sequences for Response-Consistent Interpretation of TBM Excavation Processes

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Abstract

Existing representations of rock–TBM interaction during excavation treat monitoring responses as temporal sequences, which preserve temporal dependence but lack the spatial explicitness needed to resolve where interactions occur, which machine components are implicated, and how interaction structure evolves along chainage. This study proposes a response-supervised, geometry-constrained graph sequence framework for spatially explicit representation and interpretable learning of rock–TBM interaction. The framework introduces three key innovations: (i) a heterogeneous graph-sequence representation that formalizes TSP-derived rock voxels and parameterized TBM surface nodes as spatial entities with explicit candidate interaction relations; (ii) a geometry-constrained edge construction strategy based on spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints, which restricts learning to spatially plausible rock–machine relations rather than unconstrained connectivity; and (iii) a graph-to-surface interpretation route that projects learned edge relevance onto TBM surface and chainage spaces to produce response-consistent hotspot maps and interaction-evolution views. Rather than relying on sparse jamming labels or manually calibrated contact forces, the framework uses continuously recorded monitoring responses to supervise the relative importance of geometry-screened candidate relations, enabling interpretable spatial learning under incomplete observability. The evaluation compares the framework against monitoring-only sequence models, TSP-augmented sequence models, tabular baselines, and structural graph ablations under a mileage-ordered forward evaluation. The proposed framework remains competitive with the sequence baselines but does not establish a statistically significant predictive advantage on the shifted future test interval. Checkpoint-based diagnostics nevertheless show spatially clustered surface relevance and strong inverse co-variation with TSP velocity along the test interval; these patterns are reported as response-consistent diagnostics, not as proof of physical contact or geological causality. By unifying geological sensing, machine geometry, monitoring response, and spatial interpretation within a constrained graph sequence, the framework offers a spatially explicit approach to excavation interaction analysis that complements existing temporal prediction methods.

Keywords: spatial representation; heterogeneous graph sequence; rock–machine interaction; geometry-constrained edge construction; response supervision; interpretable geospatial learning; TBM excavation; spatial interpretation

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1 Introduction

Tunnel boring machines (TBMs) are widely used in long and deep tunnel construction, where excavation responses emerge from the coupled effects of geological conditions, machine geometry, and operational process. Under fractured, water-rich, high-stress, or locally deformable ground conditions, the interaction state between surrounding rock and the TBM may change rapidly and manifest through abnormal thrust, torque, penetration, advance rate, or shield-related responses. These monitoring variables are informative external expressions of rock–TBM interaction, but monitoring curves alone provide limited evidence about where the interaction occurs, which machine components are implicated, and how local interaction relations evolve along chainage.

Recent developments in tunnel seismic prediction (TSP), advance geological prediction, digital tunnelling, and TBM monitoring systems have substantially increased the availability of heterogeneous excavation data [11]. TSP provides a spatially distributed description of the geological field ahead of or around excavation, whereas monitoring systems provide temporally continuous responses referenced to excavation progress. The challenge is therefore not only data acquisition, but also data organization: heterogeneous sources with different spatial support, temporal resolution, and physical meaning must be integrated into a representation that preserves excavation evolution together with local spatial structure.

This challenge can be viewed as a geographic information representation problem. Three GIScience concerns are central. First, *spatial entity formalisation*: TSP voxels, TBM surface components, and monitoring locations are heterogeneous spatial objects with distinct spatial supports (point, surface, volume) and semantic roles, and must be abstracted into a unified entity model that preserves their geometric and topological properties [4, 6, 9]. Second, *constrained relation construction*: candidate rock–machine relations cannot be learned freely from black-box correlation but must respect geometric compatibility, excavation state, and active-zone constraints—an instance of the broader GIScience principle that spatial relations should be constructed with domain-plausible topology rather than unconstrained connectivity [3, 15, 30]. Third, *interpretable spatial learning*: the learned representation must be projectable back to interpretable spatial views (surface hotspots, chainage evolution) to support spatial reasoning, not merely optimise a prediction loss [10, 13]. In this light, the problem extends beyond conventional engineering prediction and can be framed as a GIScience problem—one that benefits from explicit spatial entity modelling, constrained relation construction, and interpretable spatial learning rather than merely feature concatenation or temporal sequence extension.

Existing TBM prediction and warning studies have shown that tabular models and sequence models can be effective for forecasting penetration, thrust, torque, or jamming-related responses [1, 7, 25, 26]. However, most of these methods represent excavation either as a feature table or as an ordered monitoring sequence. Sequence models are effective for learning temporal dependence in monitoring records [8, 19], but they usually treat excavation as an ordered signal sequence rather than as a spatially structured interaction process. As a consequence, they have limited ability to preserve rock location, machine-component structure, and local rock–machine candidate relations within a unified representation. This limitation matters when the objective is not only response prediction, but also spatially consistent interpretation of interaction dynamics.

Graph learning offers a natural alternative because it is well suited to irregular objects and relations. Graph-based models have shown strong capacity for encoding object interaction, state propagation, and structured spatial systems in domains such as physics simulation [16, 17], urban flow prediction [5, 28], and spatio-temporal forecasting [12, 24]. Heterogeneous graph networks further enable the representation of multi-type entities and relations [18, 22, 27]. Yet TBM excavation is not a generic graph-learning problem in which edges can be learned freely from black-box correlation. Candidate rock–TBM relations must remain consistent with machine geometry, excavation state, and local active zones. In this setting, graph learning is

most meaningful when it operates on geometrically plausible candidate relations rather than on unconstrained connectivity.

Motivated by this gap, this study proposes a response-supervised, geometry-constrained graph sequence framework for spatially explicit rock–TBM interaction representation and interpretation. TSP-derived geological attributes are organized as rock voxel nodes, while the cutterhead and shield are discretized as parameterized TBM surface nodes. Candidate rock–machine edges are screened through spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints. Instead of interpreting edge weights as direct physical contact forces or relying on sparse jamming labels, the framework uses continuously recorded monitoring responses to supervise the relative importance of geometry-screened candidate relations. The learned relevance is then projected back to TBM surface space and chainage space to generate response-consistent hotspot and evolution views.

The study addresses three questions: how to organize TSP rock attributes, TBM surface structure, and rock–machine candidate relations into a dynamic graph sequence that preserves spatial explicitness and relation plausibility; how to learn the relative importance of candidate relations from continuous monitoring responses without sparse event labels, enabling interpretable spatial learning under incomplete observability; and how to map learned interaction relevance back to interpretable spatial views that support spatial reasoning beyond prediction. The main contributions are fourfold.

1. *A spatially explicit heterogeneous graph-sequence representation for rock–machine interaction.* Unlike feature-table or pure-sequence representations that discard spatial structure, the proposed representation formalises TSP voxels, TBM surface components, and their candidate relations as typed spatial entities with explicit geometric and topological properties, preserving spatial explicitness throughout the learning pipeline.
2. *A geometry-constrained relation construction strategy that embeds domain plausibility into graph topology.* Candidate rock–machine edges are screened by spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints. This ensures that the learned relations remain geometrically plausible, addressing the GIScience principle that spatial relations should be constructed with domain-plausible topology rather than unconstrained connectivity.
3. *A response-supervised relation learning paradigm that operates under incomplete observability.* Without requiring contact-force labels or sparse jamming events, the framework uses continuously recorded monitoring responses to supervise the relative importance of candidate relations, enabling interpretable spatial learning in settings where ground-truth interaction labels are unavailable.
4. *A graph-to-surface interpretation route that supports spatial reasoning beyond prediction.* Learned interaction relevance is projected back to TBM surface hotspots and chainage-evolution views, accompanied by spatial consistency indicators (Moran’s I , component CV, attention–geology correlation) that quantify the spatial plausibility of the learned representation.

2 Methodology

This section formalizes the proposed framework as a geometry-constrained heterogeneous graph sequence supervised by continuously recorded TBM monitoring response. The method represents excavation as an evolving system of spatial entities—rock units and TBM surface components—and their candidate interaction relations, rather than as a monitoring-only signal sequence. By formalizing spatial entities, constraining candidate relations by geometric compatibility, and projecting learned relevance back to interpretable spatial views, the framework

addresses the geographic-information representation problem identified in the Introduction: how to organize heterogeneous spatial entities into a representation that preserves spatial explicitness, relation plausibility, and interpretability.

2.1 Framework overview

Let excavation advance be discretized into chainage-indexed steps $t = 1, 2, \dots, T$. For each step, the framework constructs a graph snapshot G_t and then models the ordered graph sequence $\{G_t\}_{t=1}^T$. The workflow contains three stages: geometry-constrained graph-sequence construction, response-supervised interaction learning, and graph-to-surface spatial interpretation (Figure 1). Given the most recent K excavation steps, the model uses graph snapshots together with synchronized TBM monitoring vectors to predict a future response at step $t + h$:

$$\hat{r}_{t+h} = f_{\Theta}(G_{t-K+1:t}, u_{t-K+1:t}),$$

where u_t denotes the monitoring vector at step t , r_{t+h} denotes the target future response, K is the historical window length, and h is the forecasting horizon. In this study, the learned rock-machine edge attention is interpreted as response-consistent interaction relevance under the forecasting task, not as true contact force, contact pressure, jamming probability, or any direct engineering risk quantity.

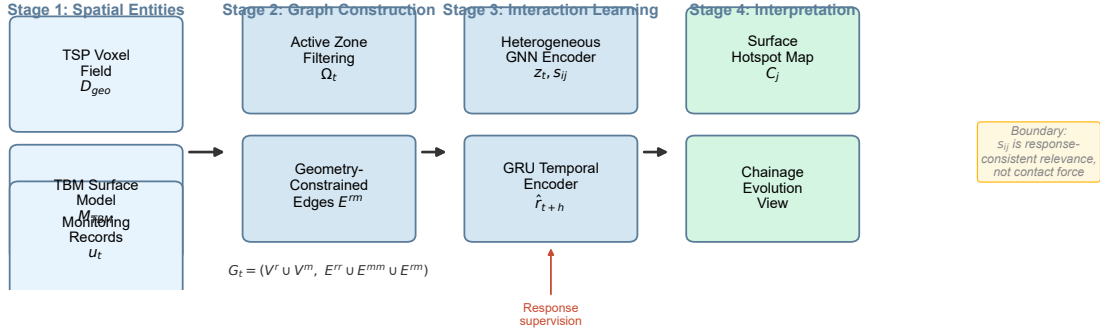


Figure 1: Framework pipeline (schematic). Node counts and edge counts are downsampled for visual clarity; the actual graph uses 1352 TBM surface nodes and approximately 3500 active-zone rock nodes per step (see Implementation details). Stage 1 constructs spatial entities from the TSP voxel field, TBM surface geometry, and monitoring records. Stage 2 builds geometry-constrained graph snapshots through active-zone filtering and candidate rock-machine edge construction. Stage 3 performs response-supervised graph-sequence learning using a heterogeneous GNN encoder and GRU temporal encoder. Stage 4 projects learned relevance back to TBM surface and chainage spaces to generate hotspot maps and evolution views.

2.2 Excavation-step definition and data alignment

Raw TBM monitoring records are usually sampled at a much higher frequency than the graph sequence should use. The method therefore first aggregates monitoring data by chainage or ring-based excavation progress to form a unified excavation-step index t . A practical initial aggregation granularity is 0.05 m or 0.1 m, subject to data density and desired spatial resolution. Each excavation step is associated with four objects: current face or shield position, graph snapshot G_t , synchronized monitoring vector u_t , and future response label r_{t+h} .

The monitoring input vector can include

$$u_t = [\text{AdvanceRate}_t, \text{RPM}_t, \text{Torque}_t, \text{Thrust}_t, \text{Penetration}_t, \text{ShieldPressure}_t],$$

while the prediction target can be defined as a multivariate future response vector

$$r_{t+h} = [\text{AdvanceRate}_{t+h}, \text{Torque}_{t+h}, \text{Thrust}_{t+h}, \text{Penetration}_{t+h}, \text{ShieldPressure}_{t+h}].$$

Because TSP coordinates and TBM shield mileage are usually recorded in different local reference systems, the method also requires a local offset mapping. A simple alignment is to define

$$x_{\text{local}}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{\text{start}},$$

and then map $x_{\text{local}}(t)$ into the TSP coordinate range used for voxelized geological representation.

2.3 Geological and machine inputs

The geological input is a TSP-derived voxel field $D_{\text{geo}} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ denotes the spatial coordinate of rock voxel i , and g_i denotes its geological attribute vector. Depending on the available inversion and interpretation products, g_i may include seismic velocity, elastic indicators, ratio features, or other derived geological descriptors.

The TBM input is represented as a parameterized surface model $M_{\text{TBM}} = \{p_j(0), n_j, \rho_j\}$, where $p_j(0)$ is the initial coordinate of TBM surface node j , n_j is the local surface normal, and ρ_j is a machine component label such as cutterhead or shield. The excavation process is further described by advance direction d , step length Δx , step-dependent active zone Ω_t , interaction distance threshold τ , and normal-compatibility threshold $\eta_{\min}$. When a detailed three-dimensional TBM model is unavailable, the surface can be approximated by a parameterized geometry in which the cutterhead is represented as a disc and the shield as cylindrical surface components. In the current formulation, the surface is partitioned into cutterhead, front shield, middle shield, and tail shield, and nodes are sampled at fixed radial, circumferential, and axial resolutions.

2.4 Graph-sequence construction

At each excavation step t , a graph snapshot is defined as

$$G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}),$$

where V_t^r and V_t^m are rock and TBM node sets, respectively, E_t^{rr} denotes rock–rock adjacency, E_t^{mm} denotes TBM-surface adjacency, and E_t^{rm} denotes candidate rock–machine interaction edges.

Rock node features are defined as

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether a voxel remains active and eligible for interaction at step t . TBM node features are defined as

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

With constant advance step length, TBM surface nodes are updated along the advance direction by

$$p_j(t+1) = p_j(t) + \Delta x d.$$

For intra-rock connectivity, E_t^{rr} is constructed from voxel adjacency, for example a 26-neighbourhood relation. For TBM-surface connectivity, E_t^{mm} is constructed from local surface neighbourhood relations inherited from the discretized machine geometry. The key design choice concerns the candidate interaction edges E_t^{rm} , which are not fully connected but screened by geometric plausibility. This constrained construction contrasts with unconstrained graph learning [23, 29] and reflects the domain requirement that candidate rock–machine relations must be physically plausible.

2.5 Geometry-constrained candidate interaction edges

A rock voxel i and TBM surface node j are linked at step t only if the candidate relation satisfies a geometry-constrained rule:

$$e_{ij}^{rm}(t) = 1$$

if

$$\begin{cases} d_{ij}(t) \leq \tau, \\ \kappa_{ij}(t) \geq \eta_{\min}, \\ c_i \in \Omega_t, \\ s_i(t) = 1, \end{cases}$$

and the edge is absent otherwise. Here,

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

is the rock-to-surface distance, and

$$\kappa_{ij}(t) = \max \left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon} \right)$$

measures signed-normal compatibility between the TBM surface orientation and the rock-voxel direction. The active-zone condition restricts candidate relations to the geological region currently relevant to excavation, while the state variable $s_i(t)$ prevents already excavated or inactive voxels from repeatedly contributing to later interaction snapshots.

This design intentionally limits graph learning to a screened candidate set. The edge relevance learned later in the model is therefore interpreted as the relative importance of geometrically plausible candidate interactions, rather than as a direct estimate of physical contact force or stress.

2.6 Edge features and geometric prior

Each candidate rock-machine edge is further associated with an edge-feature vector

$$a_{ij}^{rm} = [d_{ij}/\tau, \kappa_{ij}, (c_i - p_j)/\tau, g_i, \text{onehot}(\rho_j)],$$

which combines three kinds of information: geometric compatibility, local geological context, and machine-component semantics. In addition, a simple geometric prior is defined as

$$\pi_{ij}^{rm} = \exp(-d_{ij}/\tau) \kappa_{ij}.$$

This prior increases when the rock voxel is closer to the TBM surface and more consistent with the local outward normal. It is used only as geometric guidance for learning and should not be interpreted as contact force, contact pressure, or a direct risk indicator. The full graph-sequence construction procedure is summarised in Algorithm 1.

2.7 Response-supervised interaction learning

After graph construction, each snapshot is encoded by a graph encoder to produce an interaction-aware latent representation. Message passing integrates geological attributes, machine-surface context, and candidate interaction structure across the heterogeneous graph. The resulting stepwise graph embeddings are then passed to a temporal encoder, which models graph-state evolution along chainage and links the sequence of graph states to continuously recorded TBM monitoring response.

For each graph snapshot, the graph encoder computes raw (pre-softmax) edge scores $s_{ij}^{rm}(t)$ for each candidate rock–machine edge, which are then normalised over all source rock nodes incident to each destination TBM node j to obtain attention weights $\alpha_{ij}^{rm}(t) = \exp(s_{ij}^{rm}(t)) / \sum_{i'} \exp(s_{i'j}^{rm}(t))$ used in message passing [20, 21]. The encoder also produces a graph-level representation z_t . The graph-level representation is concatenated with the synchronized monitoring vector u_t and passed to a gated recurrent unit (GRU) [2], which models graph-sequence evolution across consecutive excavation steps.

The supervision signal is provided by monitoring response variables recorded during excavation, rather than by sparse binary jamming labels or manually assigned edge weights. Under this design, the model learns which screened candidate rock–machine relations are most relevant for explaining or anticipating response variation. The resulting raw scores $s_{ij}^{rm}(t)$ are therefore response-consistent relevance indicators under geometric screening. The training procedure is summarised in Algorithm 2.

2.8 Graph-to-surface spatial interpretation

To recover spatial interpretability, the learned relevance associated with candidate rock–machine relations is projected back onto two explicit spaces.

Notation. We distinguish three quantities in the interpretation pipeline:

- *Raw edge score* $s_{ij}^{rm}(t)$: the pre-softmax attention score computed by the graph encoder for candidate edge (i, j) at step t . This quantity preserves absolute magnitude differences across edges and is used for all spatial interpretation outputs.
- *Normalised attention weight* $\alpha_{ij}^{rm}(t)$: the post-softmax weight used in message passing, obtained by normalising raw scores over all source rock nodes incident to destination TBM node j :

$$\alpha_{ij}^{rm}(t) = \frac{\exp(s_{ij}^{rm}(t))}{\sum_{i': (i', j) \in E_t^{rm}} \exp(s_{i'j}^{rm}(t))}.$$

Because $\sum_{i: (i, j) \in E_t^{rm}} \alpha_{ij}^{rm}(t) = 1$ for each TBM node j , aggregating α_{ij}^{rm} per node yields a constant and is therefore unsuitable for spatial interpretation.

- *Surface relevance* $C_j(t)$: the degree-normalised aggregated non-negative-transformed raw score incident to TBM surface node j , defined as

$$C_j(t) = \frac{1}{d_j(t) + \epsilon} \sum_{i: (i, j) \in E_t^{rm}} \text{softplus}(s_{ij}^{rm}(t)),$$

where $d_j(t) = |\{i : (i, j) \in E_t^{rm}\}|$ is the degree of TBM node j in the rock–machine sub-graph at step t , $\epsilon = 10^{-8}$ avoids division by zero, and $\text{softplus}(x) = \log(1 + e^x)$ is a smooth non-negative transform. The softplus is introduced because raw pre-softmax scores s_{ij}^{rm} can be negative, and direct summation would allow positive and negative scores to cancel, producing ambiguous C_j values that confound spatial interpretation. Softplus preserves the monotonic ordering of raw scores while ensuring non-negativity, so that C_j faithfully reflects the aggregate interaction relevance incident to each surface node. The degree normalisation prevents nodes with more incident candidate edges from receiving artificially inflated relevance solely due to edge count, ensuring that C_j reflects the *average per-edge relevance* rather than the raw sum. No further per-chainage standardisation (e.g. z-scoring) is applied to C_j , so that absolute magnitude differences across chainages are preserved for downstream spatial consistency indicators (Moran’s I , component CV, and attention–geology correlation all require non-degenerate per-chainage means and variances). For hotspot visualisation only, C_j is min–max rescaled per chainage for colour

mapping; the underlying values used in all quantitative indicators remain the unstandardised degree-normalised scores.

Projection. First, $C_j(t)$ is projected onto the cutterhead and shield surface to produce component-aware hotspot maps. Second, the sequence of $C_j(t)$ values is tracked along excavation chainage to visualize how interaction emphasis evolves over space and excavation progress. These outputs show where the model places interaction importance and how that importance migrates during excavation, while remaining consistent with the monitored response signal used for supervision. The interpretation procedure is summarised in Algorithm 3.

2.9 Implementation details

The framework is implemented in Python using PyTorch and PyTorch Geometric. Key hyperparameters and structural settings are summarised below; the same values are used in the Experimental setup (Section 3.6) and all ablations.

Graph structure. The TBM surface is discretised into 1352 nodes (cutterhead disc + front, middle, and tail shield cylinders) at a surface resolution of 0.5 m. The TSP voxel field yields approximately 3500 active-zone rock nodes per excavation step (filtered by the component-specific active zone: cutterhead look-ahead 5.0 m with radial extent $\tau_{\text{zone}} = 5.0$ m; shield annular ring with $R_s \leq r \leq R_s + \tau_{\text{zone}}$). Candidate rock-machine edges are constructed with distance threshold $\tau_{\text{edge}} = 2.0$ m and signed-normal compatibility threshold $\eta_{\text{min}} = 0.3$. The historical window length is $K = 5$ and the forecasting horizon is $h = 1$.

Model architecture. The heterogeneous GNN encoder uses $\text{gnn_layers} = 2$ message-passing layers with rock-node hidden dimension 32, TBM-node hidden dimension 32, and edge hidden dimension 32; dropout is 0.3. The GRU temporal encoder has a hidden dimension of 64 with 1 layer. The prediction head is a two-layer MLP with ReLU activation. Baseline LSTM and TSP-LSTM models use hidden dimension 128, 2 layers, and dropout 0.1 for comparability.

Training. All models are trained for up to 80 epochs using AdamW optimiser with initial learning rate 10^{-3} , weight decay 10^{-2} , standardised Huber loss ($\delta = 1.0$), and ReduceLROnPlateau scheduler (factor 0.5, patience 10). Batch size is 32. Early stopping with patience 15 is applied on validation loss. Monitoring variables and TSP attributes are standardised using training-set statistics only. Monitoring scalars are fitted from rows used by the chronological training samples, and TSP attribute scalars are fitted from voxels in the neighbourhood of training target chainages. Random seed is fixed at 42 for reproducibility.

Spatial consistency indicators. Moran’s I is computed using inverse-distance weights on TBM surface node positions within each chainage. Component CV is the coefficient of variation of mean C_j across the four TBM components (cutterhead, front, middle, tail shield). Attention-geology correlation is the Pearson and Spearman correlation between mean C_j and TSP V_p across test chainages.

2.10 Evaluation design

The experimental design compares the proposed framework against several baseline families: monitoring-only sequence models, TSP-augmented sequence models, tabular baselines, and graph-structure ablations. Structural ablations include randomized rock-machine edges and versions without geometric constraints. This design isolates the contribution of graph structure, geological attributes, temporal graph evolution, and geometry-based candidate screening.

The formal evaluation uses a chronological mileage-ordered split. The first 70% of prediction-target samples are used for training, the next 15% for validation, and the remaining 15% for testing. This design evaluates forward generalisation to a later continuous tunnel interval and deliberately retains the geological distribution shift encountered by the advancing TBM.

2.11 Algorithmic summary

The three stages of the framework are summarised in Algorithms 1–3.

Algorithm 1 Geometry-constrained graph-sequence construction

Require: Rock voxel field D_{geo} , TBM surface M_{TBM} , thresholds τ , $\eta_{\min}$, active-zone Ω_t

- 1: **for** each excavation step $t = 1, \dots, T$ **do**
- 2: Update TBM surface positions: $p_j(t) \leftarrow p_j(t-1) + \Delta x d$
- 3: Build rock–rock edges E_t^{rr} from 26-neighbourhood
- 4: Build TBM–TBM edges E_t^{mm} from surface adjacency
- 5: Build candidate rock–machine edges:

$$E_t^{rm} \leftarrow \{(i, j) : d_{ij}(t) \leq \tau, \kappa_{ij}(t) \geq \eta_{\min}, c_i \in \Omega_t, s_i(t) = 1\}$$

- 6: Compute edge features a_{ij}^{rm} and geometric prior π_{ij}^{rm}
 - 7: Assemble graph snapshot $G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm})$
 - 8: **end for**
 - 9: **return** Graph sequence $\{G_t\}_{t=1}^T$
-

Algorithm 2 Response-supervised interaction learning

Require: Graph sequence $\{G_t\}$, monitoring vectors $\{u_t\}$, window K , horizon h

- 1: **for** each training step **do**
 - 2: **for** each sample window $t - K + 1$ to t **do**
 - 3: Encode each G_s by heterogeneous GNN: $z_s, \{s_{ij}^{rm}(s)\} \leftarrow \text{GNN}(G_s)$
 - 4: Concatenate graph and monitoring: $h_s = [z_s; u_s]$
 - 5: **end for**
 - 6: Model temporal evolution: $\{h_s\} \leftarrow \text{GRU}(\{h_s\})$
 - 7: Predict: $\hat{r}_{t+h} = \text{FC}(h_T)$
 - 8: Compute loss: $\mathcal{L} = \text{Huber}(\hat{r}_{t+h}, r_{t+h})$
 - 9: Update parameters Θ by back-propagation
 - 10: **end for**
 - 11: **return** Trained model f_{Θ} , raw edge scores $\{s_{ij}^{rm}(t)\}$
-

Algorithm 3 Graph-to-surface spatial interpretation

Require: Trained model f_Θ , test graph sequence, TBM surface

- 1: **for** each test chainage c **do**
 - 2: Extract pre-softmax edge scores $s_{ij}^{rm}(c)$ for rock-machine edges
 - 3: Aggregate per TBM node (degree-normalised): $C_j(c) = \frac{1}{d_j(c)+\epsilon} \sum_{i:(i,j) \in E^{rm}} \text{softplus}(s_{ij}^{rm}(c))$
 - 4: Project $C_j(c)$ onto TBM surface $\rightarrow$ hotspot map at chainage c
 - 5: **end for**
 - 6: Track $C_j(c)$ along chainage $\rightarrow$ component-wise evolution view
 - 7: **Output:** Hotspot maps, chainage-evolution view
-

3 Experiments and results

3.1 Study area

The study area is a shield TBM tunnelling project in southwestern China. The tunnel traverses a geologically diverse section comprising intact limestone, fractured zones, and transition regions, as revealed by the tunnel seismic prediction (TSP) survey. The TSP-forecast seismic velocity field exhibits substantial spatial heterogeneity along the tunnel alignment, with P-wave velocities ranging from 3 976 to 4 801 m/s (mean 4 429 m/s, SD = 144 m/s), reflecting variations in rock mass quality from weak fractured zones to competent limestone. The graph-construction procedure is illustrated schematically in Figure 2.

3.2 Monitoring data

The TBM monitoring system records six operational variables at each excavation step (1-m chainage interval): advance rate (mean 43.8 mm/min, range 19.4–60.3), cutterhead RPM (mean 6.0 rev/min, range 5.4–7.0), torque (mean 3 265 kN·m, range 2 253–4 484), thrust (mean 19 302 kN, range 13 300–26 761), penetration (mean 11.7 mm/rev, range 3.9–17.5), and shield pressure (mean 1.67 bar, range 1.09–2.55). These variables collectively characterise the machine’s mechanical response to the surrounding rock mass and serve as both input features (historical window) and prediction targets (forecasting horizon).

3.3 Geological forecast data

The TSP survey provides ahead-of-face geological forecasts, including seismic P-wave velocity (V_p), S-wave velocity (V_s), density (ρ), dynamic Young’s modulus (E), velocity ratio (V_p/V_s), and Poisson’s ratio (ν), sampled on a 1-m voxel grid in the tunnel vicinity. The TSP data comprises 21 609 voxels covering a spatial extent of approximately $48 \times 20 \times 20$ m around the tunnel alignment. After chainage-based aggregation, the monitoring and TSP data are aligned to 49 raw excavation steps (1-m chainage interval, chainage 0–48 m).

3.4 Data partitioning

The 49 raw excavation steps form the basis for sequence sample construction. With a historical window of $K = 5$ steps and a forecasting horizon of $h = 1$ step, each sequence sample uses 5 consecutive steps as input and predicts the response at the 6th step. This yields $49 - 5 - 1 + 1 = 44$ effective sequence samples. Samples are divided in chainage order according to their prediction targets: 30 training samples (targets at 5–34 m), 6 validation samples (35–40 m), and 8 test samples (41–48 m). This mileage-ordered split preserves the forward prediction setting and prevents later target responses from entering model fitting or preprocessing.

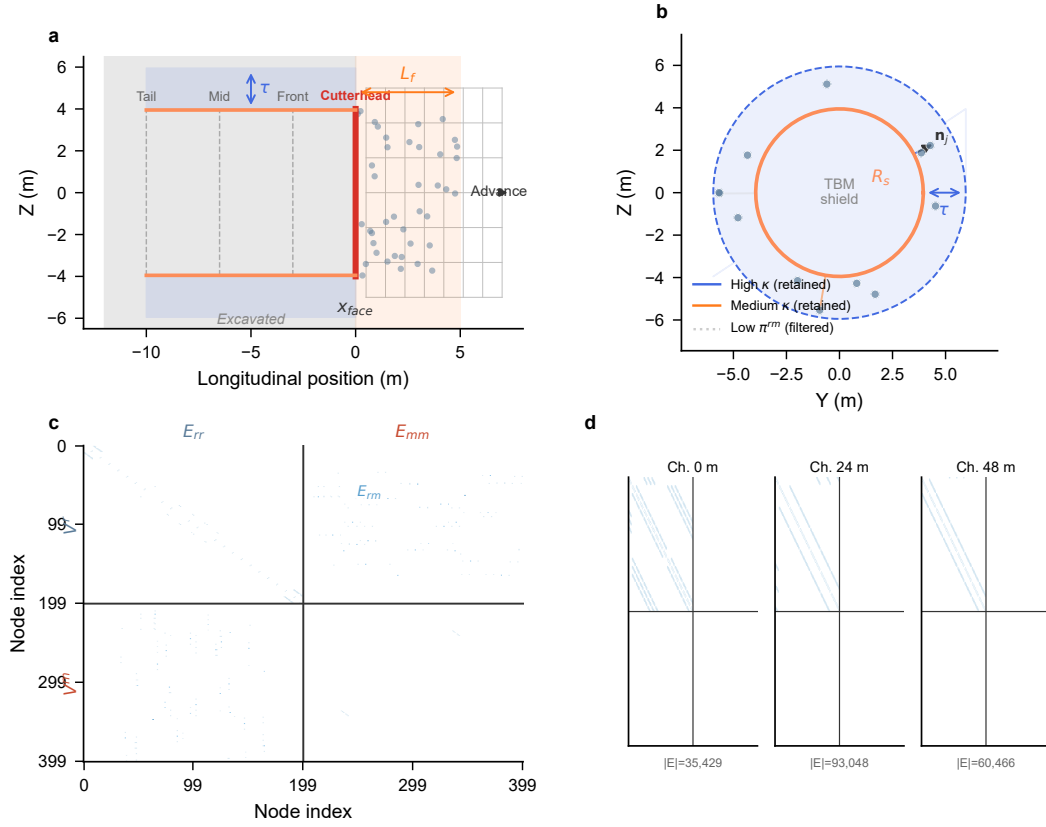


Figure 2: Schematic of geometry-constrained graph construction at a representative excavation step (illustrative, not drawn from the actual TSP velocity field). (a) Conceptual longitudinal section indicating spatial heterogeneity along the tunnel alignment, with weak fractured zones (low V_p) and competent limestone (high V_p) shown schematically; (b) active-zone definition (semi-transparent box ahead of and around the TBM) and geometry-constrained rock-machine candidate edge screening, where only rock voxels within the active zone and satisfying distance, normal-compatibility, and excavation-state constraints are connected to TBM surface nodes; (c) resulting heterogeneous graph snapshot with rock voxels (blue), TBM surface nodes (orange, coloured by component: cutterhead, front, middle, tail shield), and geometry-constrained rock-machine edges (grey).

Input windows at partition boundaries may include earlier chainages from the preceding partition, as required by a rolling forecast. For example, the validation sample targeting 35 m uses observations from 30–34 m. No monitoring response at or after a sample’s target chainage is used in that sample’s input. The TSP forecast is treated as ahead-of-face geological information available before excavation reaches the target chainage. Table 1 summarises the formal partition.

Table 1: Dataset summary for the mileage-ordered evaluation. Effective samples are sequence windows of length $K + h = 6$, and partitions are defined by prediction-target chainage.

Partition	Prediction-target chainage (m)	Required input chainage (m)	Effective samples
Training	5–34	0–33	30
Validation	35–40	30–39	6
Test	41–48	36–47	8
Total	–	5–48 m target range	44

3.5 Scope and interpretive boundaries

Before presenting the experimental results, three scope limitations should be noted to avoid over-interpretation. First, the learned edge relevance and surface relevance C_j reflect response-consistent interaction patterns under the forecasting task; they are *not* direct measurements of physical contact force, contact pressure, or jamming probability. Second, the case study uses a single tunnel section with 49 excavation steps (44 effective sequence samples, 8 in the forward test set); all quantitative indicators should be read as illustrative rather than statistically conclusive. Third, the spatial consistency indicators (Moran’s I , component CV, attention–geology correlation) quantify the internal and external plausibility of the learned representation, not its engineering correctness. These boundaries are revisited in the Discussion.

3.6 Experimental setup

All structural and training hyperparameters follow the Implementation details in Section 2 (summarised here for readability): historical window $K = 5$, forecasting horizon $h = 1$; candidate edge distance threshold $\tau_{\text{edge}} = 2.0$ m; signed-normal compatibility threshold $\eta_{\text{min}} = 0.3$; active-zone radial extent $\tau_{\text{zone}} = 5.0$ m; TBM surface resolution 0.5 m (1352 nodes); GNN encoder with 2 layers, hidden dimension 32, dropout 0.3; GRU hidden dimension 64, 1 layer; AdamW optimiser, learning rate 10^{-3} , weight decay 10^{-2} , batch size 32, early stopping patience 15. Samples are partitioned in chainage order into 30 training, 6 validation, and 8 test samples. All models are evaluated on the held-out future test interval using MAE, RMSE, R^2 , and Pearson correlation.

3.7 Prediction performance

Table 2 reports the global prediction performance (Figure 3). The proposed framework achieves MAE 0.435, RMSE 0.545, $R^2 = -0.463$, and Pearson $r = 0.507$. Its MAE is comparable to LSTM (0.429) and TSP-LSTM (0.436), but higher than Persistence (0.415). All models have negative R^2 on the future interval, consistent with the strong geological distribution shift between training and test chainages. Paired permutation tests find no significant MAE difference between the full framework and any baseline ($p \geq 0.393$). The results therefore do not establish a predictive advantage; the framework is evaluated primarily for the additional spatial representation and diagnostic outputs unavailable to monitoring-only models.

Table 2: Global prediction performance on the mileage-ordered future test interval ($n = 8$, target chainages 41–48 m). Negative R^2 values for all models indicate limited forward generalisation under geological shift.

Method	MAE	RMSE	R^2	Pearson r
Persistence	0.415	0.593	−0.621	0.487
XGBoost + TSP	0.505	0.679	−1.654	0.519
LSTM	0.429	0.531	−0.356	0.591
TSP-LSTM	0.436	0.536	−0.397	0.625
Proposed framework	0.435	0.545	−0.463	0.507

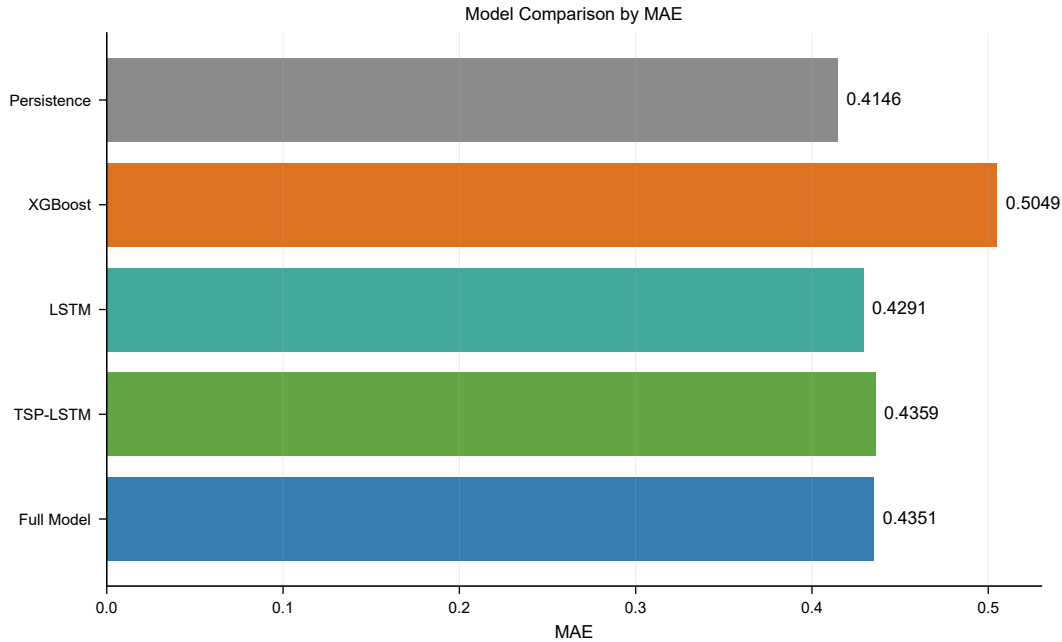


Figure 3: Global MAE comparison on the mileage-ordered future test interval. The proposed framework is competitive with the sequence baselines but does not outperform Persistence.

3.8 Structural ablation

Table 3 and Figure 4 present the structural ablation results. Four ablation variants isolate the contribution of each design component.

Table 3: Structural ablation results on the mileage-ordered future test set. Each row removes or replaces one design component from the full framework. The geometry-only baseline is excluded from this table because it has no trainable parameters and does not produce predictions; its spatial consistency is reported in Table 4.

Variant	MAE	RMSE	R^2	Pearson r
Full framework	0.435	0.545	-0.463	0.507
Without monitoring input	0.475	0.562	-0.737	0.360
Randomised rock-machine edges	0.413	0.509	-0.264	0.580
Without geometric prior ($\beta = 0$)	0.519	0.614	-0.967	-0.186

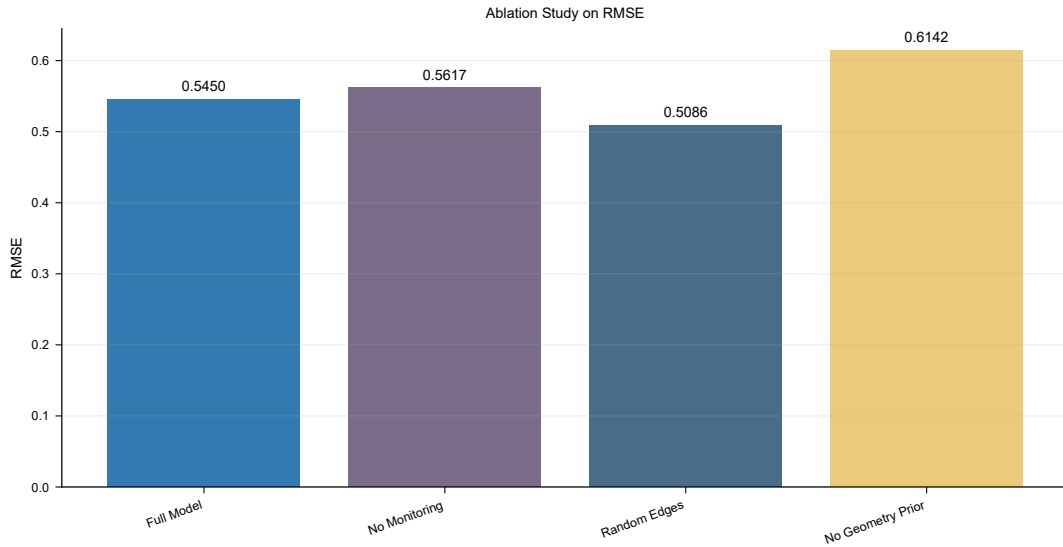


Figure 4: RMSE comparison across structural ablations on the mileage-ordered future test interval. The random-edge variant performs better than the full framework on this small shifted test set, so the ablation does not support a predictive benefit from the constrained edge topology.

The *No Monitoring* ablation removes the monitoring vector from the GRU input. Pearson r decreases from 0.507 to 0.360, suggesting that temporal monitoring dynamics carry essential predictive signal that the static graph structure alone cannot replace.

The *Randomised Edges* ablation replaces geometry-constrained rock-machine edges with random edges of the same cardinality. This variant improves MAE from 0.435 to 0.413 and Pearson r from 0.507 to 0.580. Consequently, the current forward experiment does not demonstrate that the geometry-constrained topology improves prediction.

The *No Geometric Prior* ablation sets $\beta = 0$. Its MAE increases to 0.519 and Pearson r becomes negative (-0.186), indicating that the prior remains useful within the learned topology even though the topology itself does not outperform randomised edges. Spatial consistency is examined separately below using evidence regenerated from the saved checkpoint.

The *Geometry-only baseline* replaces all learned edge scores with the fixed geometric prior $g_{ij} = \exp(-d_{ij}/\tau_{\text{edge}})$ (no trainable parameters). Because it has no learned forecasting head, it does not produce predictions and is therefore excluded from Table 3; it is, however, retained in the spatial consistency comparison (Table 4) because it yields a well-defined surface relevance

map C_j directly from the geometric prior. This baseline serves as a non-learned reference point: it provides a fixed non-learned spatial reference for evaluating whether the learned relevance refines or simply mirrors the candidate-edge geometry.

3.9 Spatial consistency of learned attention

A central claim of this framework is that geometry-constrained graph construction can produce spatially consistent attention patterns that remain inspectable in TBM surface space. We evaluate this claim using three quantitative spatial consistency indicators, each targeting a different aspect of spatial validity.

Attention–geology correlation. We quantify whether the mean surface relevance $\bar{C}(c_k) = |V^m|^{-1} \sum_j C_j(c_k)$ co-varies with the TSP P-wave velocity interpolated at the same test chainages. This check is treated as an external plausibility indicator, not as a required success criterion: in a response-supervised setting, learned relevance may also reflect machine operating state and temporal response history. The post-hoc checkpoint evidence shows strong inverse co-variation on the eight contiguous test chainages (Pearson $r = -0.988$, $p < 0.001$; Spearman $r = -1.000$). The geometry-only reference has a positive but non-significant correlation (Pearson $r = 0.549$, $p = 0.159$). Because chainage, relevance, and geological velocity all vary monotonically over this short interval, the learned correlation is treated as a local diagnostic rather than causal validation.

Spatial autocorrelation of surface relevance. Spatially consistent relevance should exhibit positive spatial autocorrelation: nearby TBM surface nodes should have similar C_j values, rather than random spatial noise. We compute Moran’s I [14] for the surface relevance values on the TBM surface at each test chainage:

$$I(c) = \frac{n}{S_0} \cdot \frac{\sum_i \sum_j w_{ij} (C_i(c) - \bar{C}(c)) (C_j(c) - \bar{C}(c))}{\sum_i (C_i(c) - \bar{C}(c))^2},$$

where $n = |V^m|$ is the number of TBM surface nodes, w_{ij} is the spatial weight between nodes i and j (binary k -nearest-neighbour, $k = 8$, row-standardised), and $S_0 = \sum_i \sum_j w_{ij}$. Moran’s I ranges from approximately -1 (perfect dispersion) through 0 (spatial randomness) to $+1$ (perfect spatial clustering). The reported value is the mean $I(c)$ across all test chainages. The full framework achieves Moran’s $I = 0.529$, indicating positively clustered surface relevance. The geometry-only baseline yields a near-zero negative mean Moran’s I (-0.054), suggesting that the learned scores are not merely a direct copy of the fixed geometric prior.

Component-level relevance differentiation. If the surface relevance captures meaningful rock–machine interaction structure, different TBM components (cutterhead, front/middle/tail shield) should exhibit distinct mean relevance levels, reflecting their different interaction intensities with the surrounding rock. We measure the coefficient of variation (CV) of component-mean relevance across the four components:

$$\text{CV} = \frac{\sigma_{\text{comp}}}{\mu_{\text{comp}}}, \quad \mu_{\text{comp}} = \frac{1}{4} \sum_{k=1}^4 \bar{C}_k, \quad \bar{C}_k = \frac{1}{|V_k^m|} \sum_{j \in V_k^m} C_j,$$

where V_k^m denotes the set of TBM surface nodes belonging to component k , and σ_{comp} is the standard deviation of the four component means. CV is dimensionless and scale-invariant, making it suitable for comparing differentiation across models with different absolute relevance magnitudes. The full framework achieves mean $\text{CV} = 0.018$, higher than the geometry-only baseline (0.006) but still small, indicating weak component-level differentiation on the future interval.

Degree-control check. Because rock-machine candidate edges are spatially constrained, nodes with more incident edges could appear more relevant simply because of edge count. We therefore correlate degree-normalised relevance C_j with incident candidate-edge degree d_j across TBM surface nodes. The mean Pearson correlation is $r = -0.225$. The negative association indicates that relevance is not a direct positive proxy for candidate-edge count, while also showing that degree normalisation does not make the two quantities fully independent.

Table 4: Post-hoc spatial evidence regenerated from the saved checkpoint. Higher Moran’s I and component CV indicate greater spatial structure. Rel.-Geo. r is the Pearson correlation between mean surface relevance and TSP P-wave velocity across test chainages. Degree-control r correlates C_j with incident candidate-edge degree and should be close to zero if relevance is not merely edge-count driven.

Variant	Rel.-Geo. r	Moran’s I	CV	Degree r
Full framework	−0.988	0.529	0.018	−0.225
Geometry-only baseline	0.549	−0.054	0.006	—

These indicators support a narrow interpretation: the saved model produces positively clustered surface relevance with a strong chainage-local inverse association with TSP velocity, but only weak differentiation among TBM components. The results do not establish physical interaction correctness or geological causality.

3.10 Spatial interpretation outputs

Figure 5 presents a response-consistent hotspot map projected onto the TBM shield surface at a future-test chainage. Colour intensity indicates the aggregated edge relevance incident to each surface node. The limited colour contrast is consistent with the small component CV and indicates weak localisation in this checkpoint.

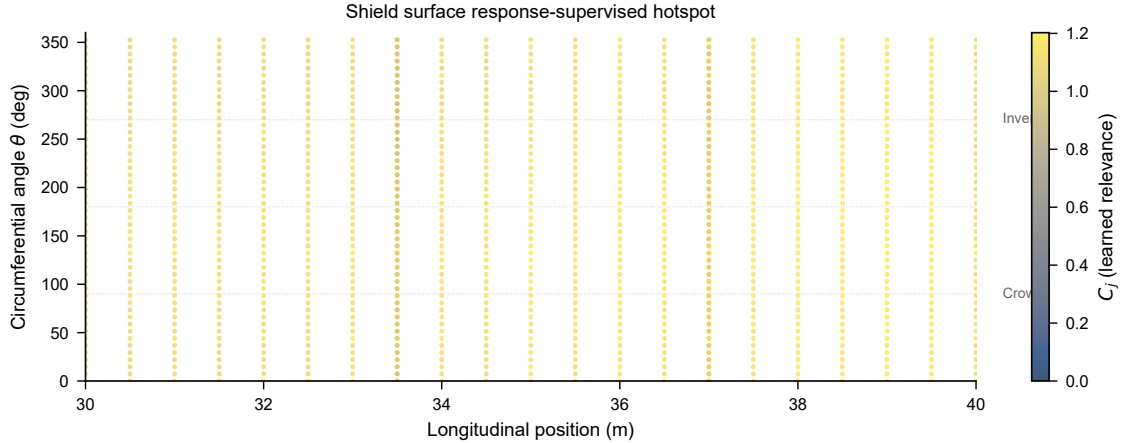


Figure 5: Response-consistent surface relevance hotspot for the sample targeting chainage 41 m, extracted from its final input graph snapshot. The colour indicates degree-normalised surface relevance C_j , rescaled within each chainage for visualisation only; different chainages are therefore not directly comparable in absolute colour intensity. The near-uniform map indicates limited hotspot localisation for this checkpoint.

Figure 6 shows the checkpoint-derived mean surface relevance across the mileage-ordered future test interval.

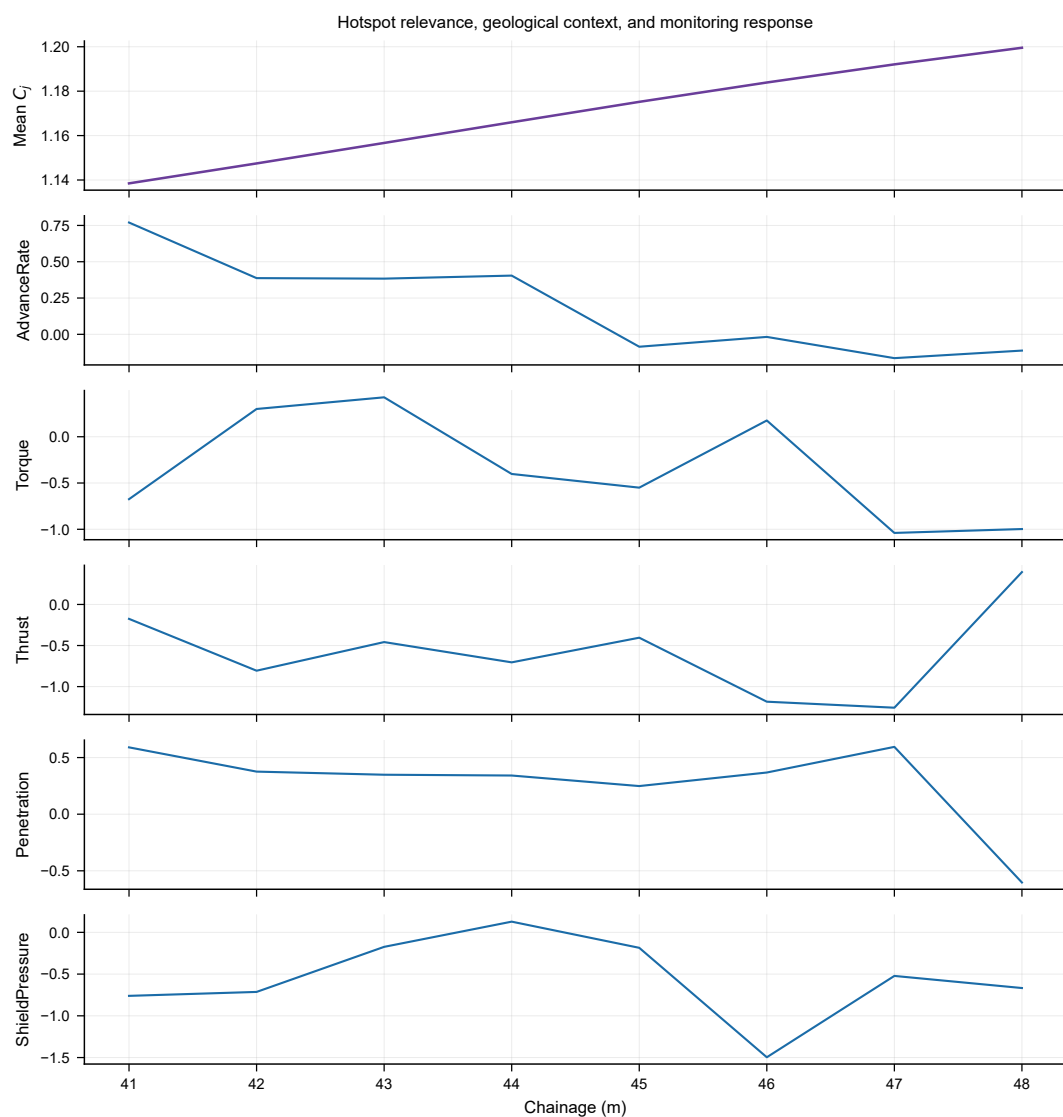


Figure 6: Checkpoint-derived relevance and monitoring responses along the contiguous future test interval (41–48 m). The top panel shows mean degree-normalised surface relevance; the remaining panels show standardised advance rate, torque, thrust, penetration, and shield pressure.

The surface map shows limited component contrast, while the chainage-indexed evolution view shows a smooth increase in mean relevance across the future test interval. These views should be interpreted as spatial diagnostics of the saved model rather than as ground-truth interaction fields. Monitoring-only sequence models cannot produce equivalent surface maps because they do not represent rock voxels, TBM surface components, or rock-machine candidate relations.

3.11 Robustness and sanity checks

Table 5 summarises the checks that are currently backed by regenerated evidence files. The reported values are produced by reloading the saved checkpoint, rebuilding the same mileage-ordered test set, and recomputing prediction and spatial evidence. Additional multi-seed, label-shuffle, and threshold-sensitivity experiments are defined in the reproducibility protocol but are not reported here until their output files are generated.

Table 5: Traceable robustness and sanity checks regenerated from the saved checkpoint. The evidence source is `outputs/evidence/posthoc_evidence.json`.

Check	Metric	Result
Checkpoint replay	MAE	0.435
Checkpoint replay	RMSE	0.545
Checkpoint replay	R^2	-0.463
Surface clustering	mean Moran’s I	0.529
Component differentiation	mean CV	0.018
Edge-count control	mean Pearson $r(C_j, d_j)$	-0.225
Geometry-only reference	mean Moran’s I	-0.054

Checkpoint replay verifies that the saved graph-sequence model reproduces the reported prediction metrics when the test set is rebuilt from the configuration file. The *edge-count control* rules out the possibility that higher C_j simply reflects a higher number of incident candidate edges. The degree-normalised definition $C_j = \frac{1}{d_j + \epsilon} \sum_i \text{softplus}(s_{ij}^{rm})$ reduces direct positive edge-count inflation in the current checkpoint. Seed stability, label permutation, and candidate-edge threshold sensitivity remain required follow-up checks before making stronger claims about robustness across training initialisations or graph construction parameters.

4 Discussion

4.1 Forward prediction under geological shift

The mileage-ordered split evaluates the model on a later continuous tunnel interval and therefore preserves the geological shift encountered during excavation. Under this setting, every model has negative R^2 . The proposed framework obtains MAE 0.435 and is statistically indistinguishable from the tested baselines. It should therefore not be presented as a superior forecasting model on the current dataset.

The ablations reinforce this cautious interpretation. Removing monitoring input degrades prediction, and removing the geometric prior produces the worst learned-model result. However, randomising the rock-machine edges improves MAE from 0.435 to 0.413. Thus the current experiment supports a contribution from monitoring history and from the prior within the chosen architecture, but does not demonstrate that the constrained edge topology improves forward prediction. The small test set and strong geological shift make this a particularly demanding evaluation.

4.2 Spatial evidence and its limits

The saved-checkpoint diagnostics show positive spatial clustering on the TBM surface (mean Moran’s $I = 0.529$). This establishes that the projected relevance is spatially structured rather than spatially random. Component-level differentiation is weak (mean $CV = 0.018$), so the current evidence does not support a strong claim that the model reliably separates cutterhead and shield interaction roles.

Mean relevance is strongly inversely associated with TSP V_p over the eight future-test chainages (Pearson $r = -0.988$, $p < 0.001$). Although the direction is consistent with stronger relevance in lower-velocity ground, this short contiguous interval also contains strong chainage trends. The correlation may therefore reflect common spatial trend rather than a direct geological response. It is retained as a local diagnostic, not as causal or physical validation.

The degree-control correlation is moderately negative ($r = -0.225$), indicating that high relevance is not simply produced by more incident candidate edges. At the same time, degree and relevance are not independent, and additional topology-shuffle and spatial-trend controls are needed before stronger interpretability claims are made.

4.3 Application scenario: spatial reasoning beyond prediction

The main value currently supported by the framework is representational. It connects geological context, monitoring response, candidate rock-machine relations, and TBM surface locations within one model and can project learned relevance back to hotspot and chainage views. Monitoring-only sequence models cannot produce an equivalent TBM-surface diagnostic because they do not encode surface nodes or rock-machine relations.

These views remain model diagnostics. They are not calibrated risk estimates, contact pressures, jamming probabilities, or prescriptions for machine control. Operational use would require independent event labels, expert assessment, or physical measurements that can validate whether the displayed hotspots correspond to actual interaction states.

4.4 Boundary and caution

The case study contains only 44 effective sequence samples and 8 future-test samples. The resulting confidence intervals are broad, and the paired tests do not establish a prediction advantage. The test interval also differs geologically from the training interval, so the negative R^2 values should be read as evidence of limited extrapolation rather than ignored as an inconvenient artifact.

The spatial evidence is checkpoint-based and post-hoc. Multi-seed stability, label permutation, candidate-edge threshold sensitivity, spatial detrending, and cross-project validation remain necessary. Learned relevance represents response-consistent model importance under geometric screening, not a direct measurement of contact force, stress, or engineering risk.

4.5 Broader GIScience implications

Despite these limitations, the framework demonstrates a GIScience-relevant representation strategy for heterogeneous spatial entities under incomplete observability. Rock voxels, TBM surface components, candidate relations, monitoring responses, and projected spatial diagnostics remain linked in one explicit entity-relation model.

The results also show why spatial graph methods require two distinct evaluation axes. Prediction metrics test forward response generalisation, whereas spatial diagnostics test whether learned relevance has coherent geographic structure. The current framework provides spatially clustered diagnostics, but neither predictive superiority nor physical validity is yet established. Keeping these claims separate is essential for a defensible interpretation.

5 Conclusion

This study proposed a response-supervised, geometry-constrained graph sequence framework for spatially explicit representation of rock–TBM interaction. The method formalises TSP voxels and TBM surface components as heterogeneous spatial entities, screens candidate relations by geometric compatibility, and projects learned relevance back to TBM surface and chainage views.

The mileage-ordered evaluation provides a direct test on a later continuous tunnel interval. The proposed framework obtains MAE 0.435, RMSE 0.545, and $R^2 = -0.463$. Its prediction performance is comparable to the sequence baselines, but no statistically significant advantage is established. The random-edge ablation performs better on the small future test set, so the current results also do not demonstrate a predictive benefit from the geometry-constrained topology.

The saved checkpoint nevertheless produces positively clustered TBM-surface relevance (mean Moran’s $I = 0.529$) and a strong inverse chainage-local association with TSP velocity (Pearson $r = -0.988$). Component differentiation is weak (mean CV = 0.018), and the geology correlation may be confounded by spatial trend. These results support the availability of spatial diagnostics, not the stronger claims of physical interaction recovery, geological causality, or operational risk estimation.

The main contribution is therefore a spatial entity-relation representation and an auditable route from response supervision to surface-based diagnostics. Future work should establish multi-seed stability, spatially detrended geology validation, topology and label permutation controls, threshold sensitivity, and cross-project forward generalisation before stronger accuracy or physical interpretability claims are made.

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Data availability statement

State where the data, code, and materials can be accessed, and note any restrictions if applicable.

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